St. Thomas' College of Engineering & Technology

Department of Artificial Intelligence & Machine Learning

CA1: Assignment

QNO. 1	Explain the concept of Linear Regression and how it is used for predicting continuous values.
QNO. 2	Discuss the difference between simple Linear Regression and Multivariate linear Regression.
QNO. 3	Provide a step-by-step procedure for implementing Linear Regression.

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Q) Explain the concepts of Linear Regression and how it is used for predicting continuous values.

→ Linear Regression is supervised machine learning Algorithm used to predict continuous numerical values.

* It establishes a linear relationship between:
  - Independent variables (x) and
  - Dependent variables (y).
  - The relationship represented using a straight line
    $y = mx + c$ [m → slope, c → intercept]
  - The model learns the best fitting line by minimizing prediction errors.
  - Errors are measured using loss functions such as Mean Square error. (MSE)
  - Once trained the model can predict unknown continuous values.

Common Example include:

- predicting house prices
- Estimating salary based on experience
- forecasting sales a temperature.

Linear Regression is fast, simple and easy to interpret. It assumes linear relationship between input and output variables.

Discuss the difference between Simple Linear Regression and Multivariate Linear regression.

Aspect	Simple linear Regression	Multivariate linear regression
Number of inputs	One independent variables.	Multiple independent variables
Equation	$y = mx + c$	$y = b_0 + b_1x_1 + b_2x_2 + \dots + b_nx_n$
Complexity	Simple and easy to interpret	More complex
Data requirement	Small datasets	Larger datasets
Use case	Single factor predicting	Real-world problems
Example	Salary vs experience	House, price vs area, rooms, location

Q No 3) Provide a step by step procedure for implementing Linear Regression

1) Define the problem:
→ Identify what needs to be predicted

2) Collect dataset:
→ Gather relevant input(x) and output(y) data.

3) Preprocess data:
→ Handle missing values and remove outliers

4) Select the project Regression model:
→ Choose simple or multivariate regression based on feature

5) Train the model:
Fit the model using training data.

6) Estimate model parameters:
Calculate slope and intercepts.

7) Make predictions:
predict output values for test data.

8) Evaluate the model:
Use metrics like MSE, RMSE, or R^2 score

9) Deploy the model:
Use trained model for real-time predictions.